Analysis of EEG Data to Investigate the Effects of Transcranial Magnetic Stimulation (TMS)

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Abstract. This report investigates the impact of transcranial magnetic stimulation (TMS) on brain activity by analyzing EEG data recorded before (preTMS) and after (postTMS) TMS administration at three intensity levels: 120%, 180%, and 240% of the lower threshold (LT). The study employs both linear and nonlinear measures to compare brain activity across conditions. Statistical tests, including ANOVA and t-tests, are used to evaluate differences in brain activity pre- and post-TMS episodes, as well as across intensity levels. The results aim to elucidate whether TMS disrupts brain function and whether this disruption is intensity-dependent.

1 Introduction

1.1 Background

Transcranial magnetic stimulation (TMS) is a non-invasive method used to modulate brain activity. While TMS is generally safe, rare cases may exhibit epileptiform discharges (EDs). This study focuses on EEG data to determine whether TMS alters brain activity and whether these alterations depend on the stimulation intensity.

1.2 Research Objectives

The primary objectives are seperated into linear and nonlinear analyses of EEG data to address the following research questions: On linear analysis:

- To compare brain activity before (preTMS) and after (postTMS) TMS administration.
- To assess whether the effects of TMS vary across different intensity levels (120%, 180%, and 240% of LT).

2 Analysis of TMS Effects on EEG Data

The approach to analyse the effects of Transcranial Magnetic Stimulation (TMS) on EEG data involved several key steps, including data selection, preprocessing, and analysis. The methodology was designed to ensure a robust and reproducible analysis of the TMS effects across different stimulation intensities. First, a pilot analysis was conducted on a single EEG channel, focusing on the effects of TMS at three different intensity levels (120%, 180%, and 240% of the lowest threshold) per pre and post segment. Afterwards, a more comprehensive linear and non linear analysis was performed on the entire dataset, which included all episodes.

2.1 Data Description

The EEG data comes from 1 patient (ID: 3) with 3 intensity levels from 60 channels, sampled at 1450 Hz. TMS was administered to the front of the skull every 2.5 sec and each recording contains a large number of TMS trials (the number is not the same in each recording). For each TMS trial an episode of 2 sec duration was extracted, where the administered single TMS (one stimulus at a time) is placed in the middle of the interval (at 1 sec). The data is organized in a 3D array (n x K x M), where $n = 2901$ (2 seconds), $K = 60$ (channels), and M is the number of TMS episodes. Each recording includes multiple TMS trials, with each trial segmented into 2-second episodes (1 second preTMS and 1 second postTMS).

2.2 Data Preprocessing and Episode Selection

In this study, we focused on a single EEG channel ('C1' notated as EEG28) among one patient. The selection of this channel was based on avoiding the referred 'bad' channels, which were excluded based on provided annotations. The preprocess was conducted on 30 episodes per intensity level. After excluding any episodes with epileptiform discharges (ED), balanced representation is ensured across patients and stimulation intensities, so 20 randomly cleaned episodes were selected per intensity using a controlled sampling strategy, resulting to 60 episodes in total. For the pilot analysis, one randomly episode was chosen per intensity level, then split into two segments of before and after the induction of TMS, leading to a total of six EEG recordings (1 seconds each). A fixed random seed (seed = 28) was set to guarantee reproducibility of results. From Patient 3, the three episodes were 7 for 120%, 9 for 180%, and 14 for 240% of the lowest threshold.

The following steps were taken:

- Bad channels were excluded based on provided annotations.
- Focus was placed on a single EEG channel for detailed pilot analysis.
- Removed any episodes with epileptiform discharges (EDs) to ensure data quality.

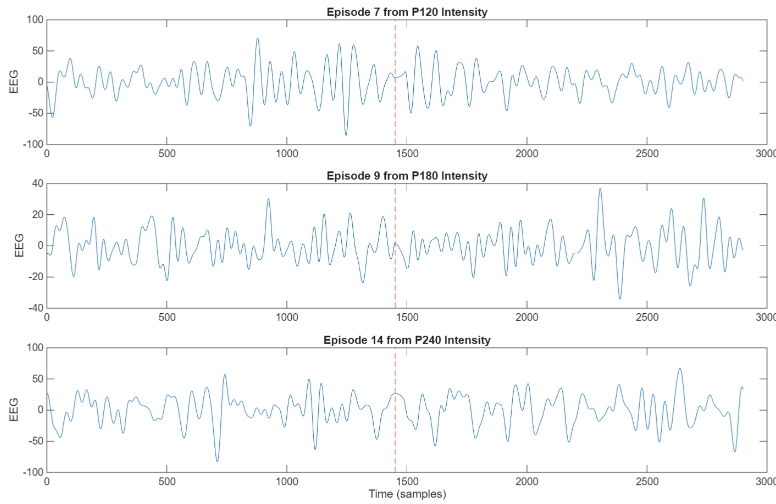

Fig. 1. Example of three TMS episodes per Intensity Level with the TMS point at 1 second. The episode is split into preTMS (0-1 sec) and postTMS (1-2 sec) segments. The red line indicates the TMS point at 1 second (1450Hz sampling rate). The episodes are from Patient 3, with three different stimulation intensities: 120%, 180%, and 240% of the lowest threshold.

– Afterwars, the pilot analysis was conducted on one randomly selected episode per intensity level and these were split on the TMS point (pre TMS and post TMS), resulting in 6 episodes for the pilot analysis.
– For later linear and non linear analysis, $N = 20$ episodes were selected per intensity level, so 60 episodes in total and 120 epochs when segmented.

2.3 Epileptiform Discharges (EDs) Exclusion

Note that in some rare cases TMS, even with a single stimulus, can cause epileptiform discharge (ED), and a characteristic waveform of short duration (a few seconds) is formed on the electroencephalogram, while the patient does not feel any effect. However, in the vast majority of TMS trials no ED is induced and after the TMS parasite the EEG appears normal, i.e. it has a similar behaviour as before the administration of TMS. This seemingly similar pattern of EEG before and after TMS does not mean that brain activity is not perturbed by TMS. The study aims to investigate just that, whether it is disturbed.

Epileptiform discharges (EDs) were identified and excluded from the analysis to ensure the integrity of the EEG data. The exclusion of EDs was based on visual inspection (this might need an automated detection method or clinical intervention), which identified episodes with significant artifacts and spikes that are not related to brain functionality or TMS induction[1]. This step was crucial to maintain the quality of the data and ensure that the analysis focused on the effects of TMS rather than confounding factors related to epileptiform activity.

2.4 Methodology Pipeline

The analysis pipeline was designed to systematically investigate the effects of TMS on EEG data in regard of brain functionality disturbance, focusing on both linear and nonlinear measures. The pilot analysis served as a preliminary step to validate the methodology and identify suitable parameters for subsequent analyses. The analysis was conducted using MATLAB with generalizability prospective with various custom scripts for linear analysis[2] and the MATS toolbox[3] for nonlinear measures. For the statistical assumptions, a customized statistical tool[4] in R was used to perform the assumption tests and the selection of the appropriate statistical tests. The pipeline consisted of several key steps, including linear and nonlinear analyses, model selections, diagnostic tests for fitted models, selection of suitable linear/ non-linear measures and statistical testing among categories after assumption testing. The following sections outline the specific procedures used in the pilot analysis and subsequent analyses.

3 Linear Pilot Analysis

After performing the data preprocessing and episodes selection, the linear pilot analysis was conducted to assess the effects of TMS on EEG data on six EEG recordings. The analysis focused on the autocorrelation and partial autocorrelation functions, model selection using ARMA models, and evaluation of the predictions using normalized root mean square error (NRMSE) in order to respond to compare the behavior between intensities and before and after the induction of TMS for the overall statistical test on the whole dataset. In short, the following steps were taken:

1. Random episode sampling procedure across the three stimulation intensities on clean episodes (already extracted EDs).
2. Each episode was split into preTMS and postTMS segments, resulting in six recordings for the pilot analysis.
3. Autocorrelation and partial autocorrelation functions were computed for a sufficient number of lags ($\tau = 60$ lags).
4. The most suitable linear ARMA models were selected based on Akaike information criterion and Ljung-Box for residual analysis (white noise criterion).
5. The models were fitted to the first 600 miliseconds (training set), evaluated on the last 400 miliseconds (validation set) and predictions were made up to ten time steps.
6. The normalized root mean square error (NRMSE) was calculated for these predictions and selected as a linear measure for statistical tests later on.

3.1 Autocorrelation and Partial Autocorrelation for the pilot analysis

The six EEG recordings were analyzed to assess the autocorrelation and partial autocorrelation functions. The analysis was performed on the preTMS and postTMS segments for each intensity level (120%, 180%, and 240% of LT). These were tested on different lags ($\tau = 40, 60, 100$ lags) to determine the sufficient statistical important lag of the timeseries to be 60 lags. The results were visualized using plots to illustrate the autocorrelation and partial autocorrelation functions for each segment.

3.2 Grid Search for ARMA Model Selection

The linear analysis involved the selection of an appropriate Autoregressive Moving Average (ARMA) model for each episode. The ARMA model is a widely used statistical model for time series analysis, which combines both autoregressive and moving average components to capture the temporal dependencies in the data. A grid search was performed to identify the best ARMA models for each episode using Akaike Criterion Information for comparison. The search involved evaluating combinations of autoregressive (AR) and moving average (MA) parameters, specifically:

- Autoregressive (AR) parameters: $p = 0, 1, 2, 3, 4, 5, 6$
- Moving Average (MA) parameters: $q = 0, 1, 2, 3, 4, 5, 6$

The best ARMA model was determined by minimizing the AIC value, ensuring that the model was both parsimonious and effective in capturing the underlying dynamics of the EEG data. In many cases, simpler orders of ARMA models were sufficient to capture the dynamics of the data as seen in the figure 3. Indicatively, we show the results for the 180% intensity level, where the ARMA models were fitted to the preTMS and postTMS segments of the EEG recordings. The aim was to avoid overfitting when the AIC values are similar to lower orders. In conjunction with the grid search procedure, the residuals of the fitted ARMA models were evaluated for whiteness using the Portmanteau test (Ljung–Box). Diagnostic plots of the residuals were also examined to assess model adequacy across each epoch category. This included verifying the sufficiency of the number of lags used in the Ljung–Box test, identifying any significant rejections, and assessing the normality of the residuals.

3.3 Model Fitting and Evaluation

As shown in the table 1, the optimal ARMA models were selected based on the lowest AIC values and the results of the Ljung-Box test. The selected models were fitted to the preTMS and postTMS segments of each episode, ensuring that the residuals met the white noise criterion. The fitted models were then used to make predictions for each episode, with a focus on evaluating the model's performance in terms of prediction accuracy.

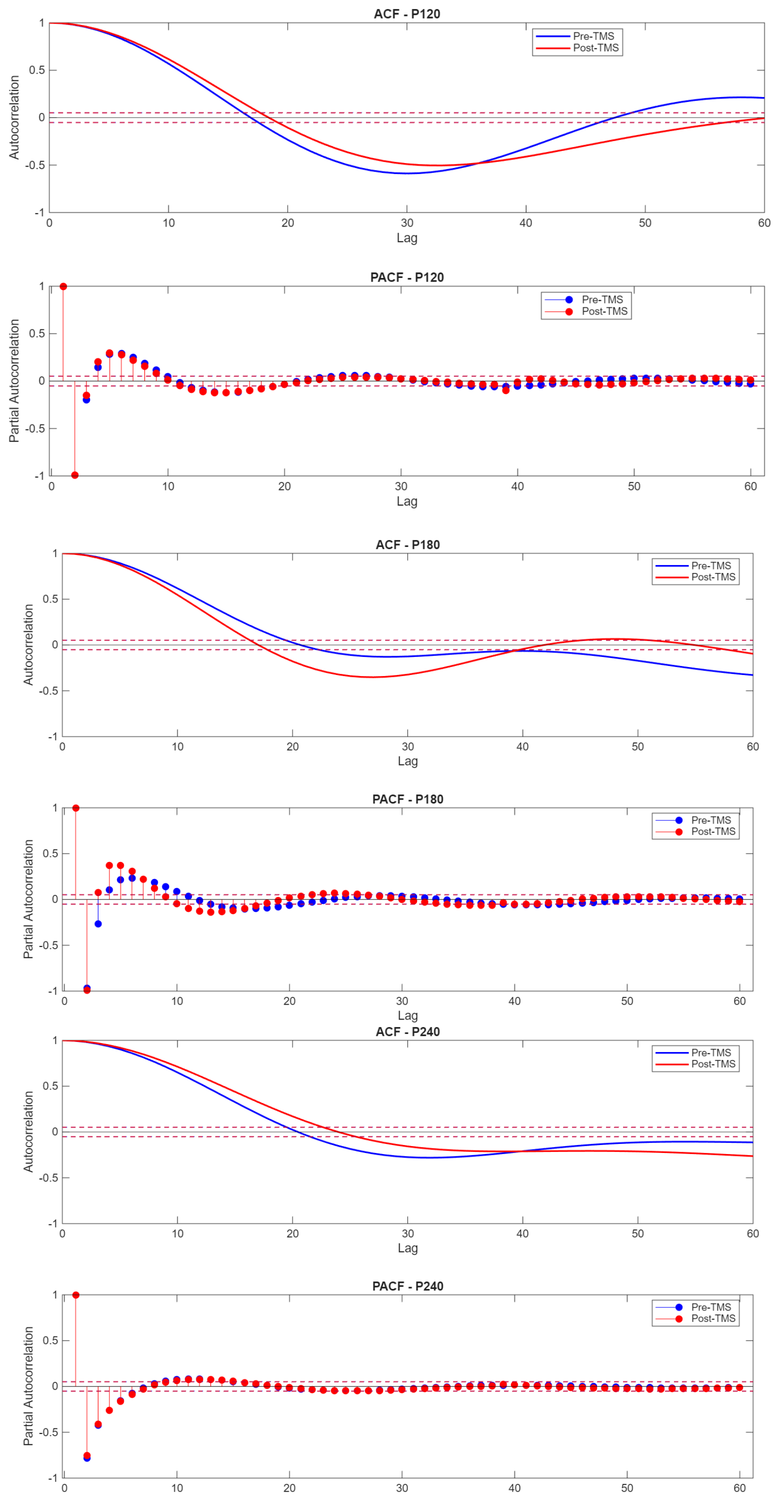

Fig. 2. Autocorrelation and Partial Autocorrelation Functions for the preTMS and postTMS segments of the six EEG recordings across three TMS intensities (120%, 180%, and 240% of LT). The plots illustrate the temporal dependencies in the EEG data before and after TMS induction.

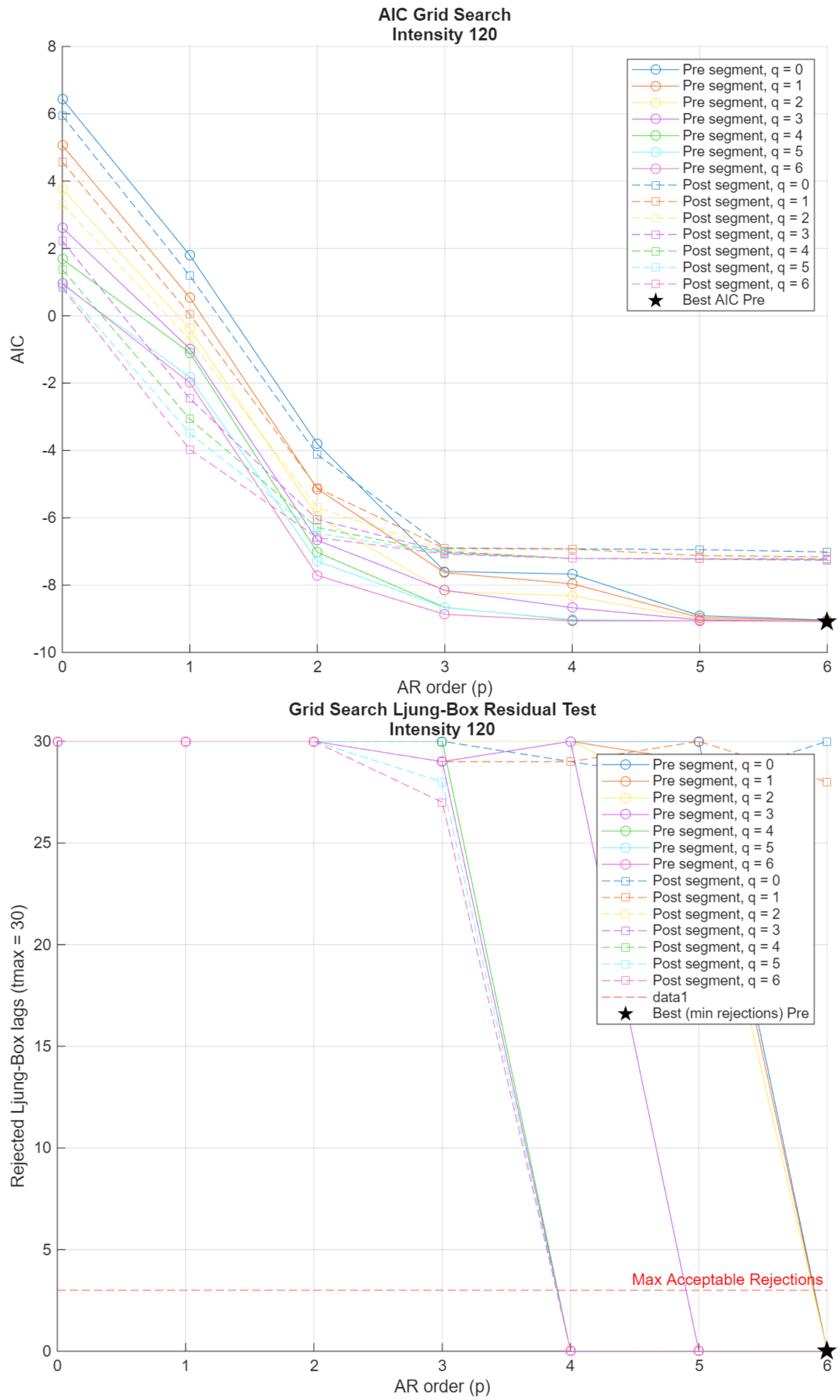

Fig. 3. Grid Search for ARMA Model Selection for EEG recordings for 120% Intensity: The plot illustrates the AIC values for different combinations of AR and MA parameters across the six EEG recordings. The optimal ARMA model is selected based on the lowest AIC value, ensuring a balance between model complexity and fit to the data. The second figure shows the residuals of the fitted ARMA models for each episode, with the Ljung-Box test results indicating whether the residuals exhibit white noise characteristics.

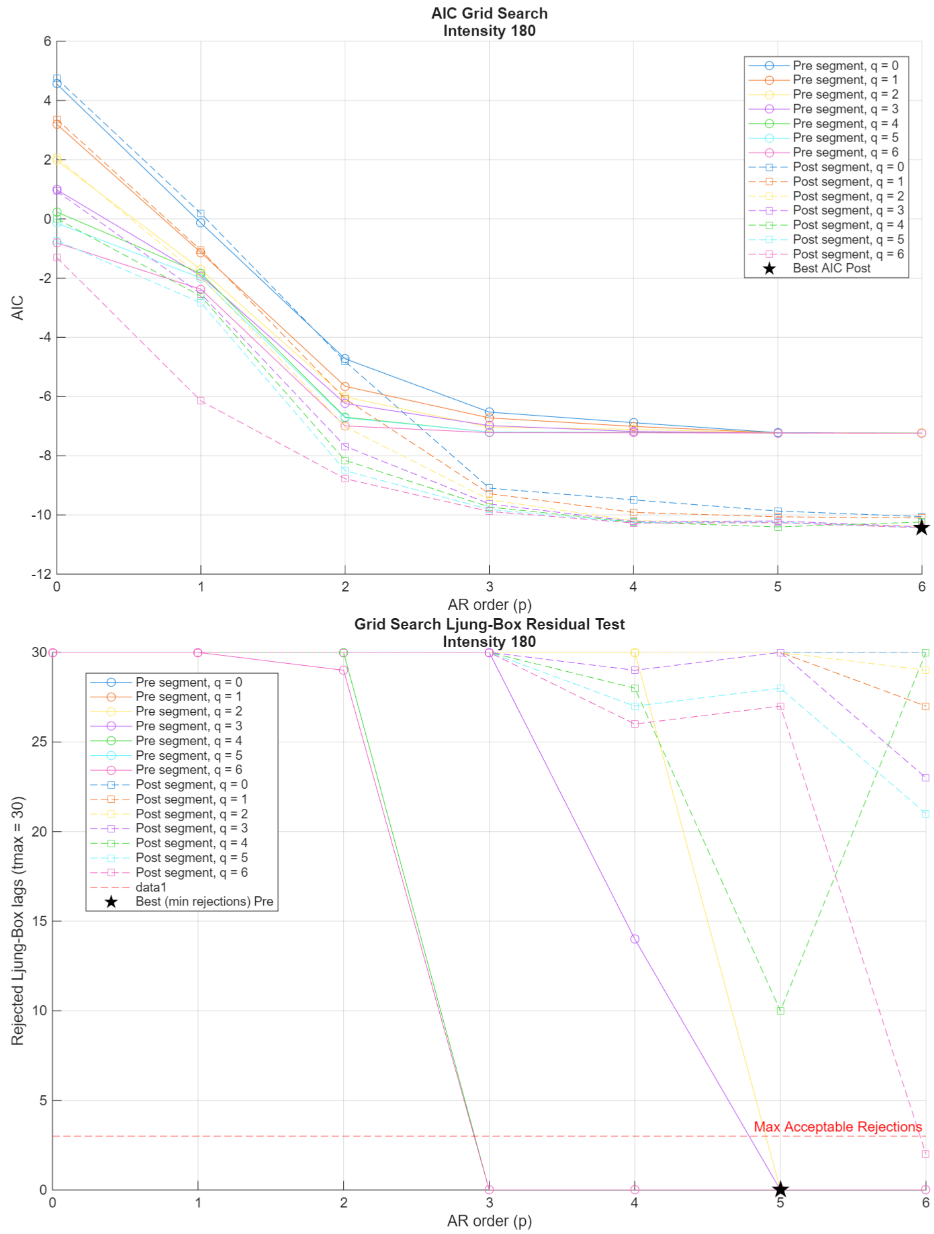

Fig. 4. Grid Search for ARMA Model Selection for EEG recordings for 180% Intensity: The plot illustrates the AIC values for different combinations of AR and MA parameters across the six EEG recordings. The optimal ARMA model is selected based on the lowest AIC value, ensuring a balance between model complexity and fit to the data. The second figure shows the residuals of the fitted ARMA models for each episode, with the Ljung-Box test results indicating whether the residuals exhibit white noise characteristics.

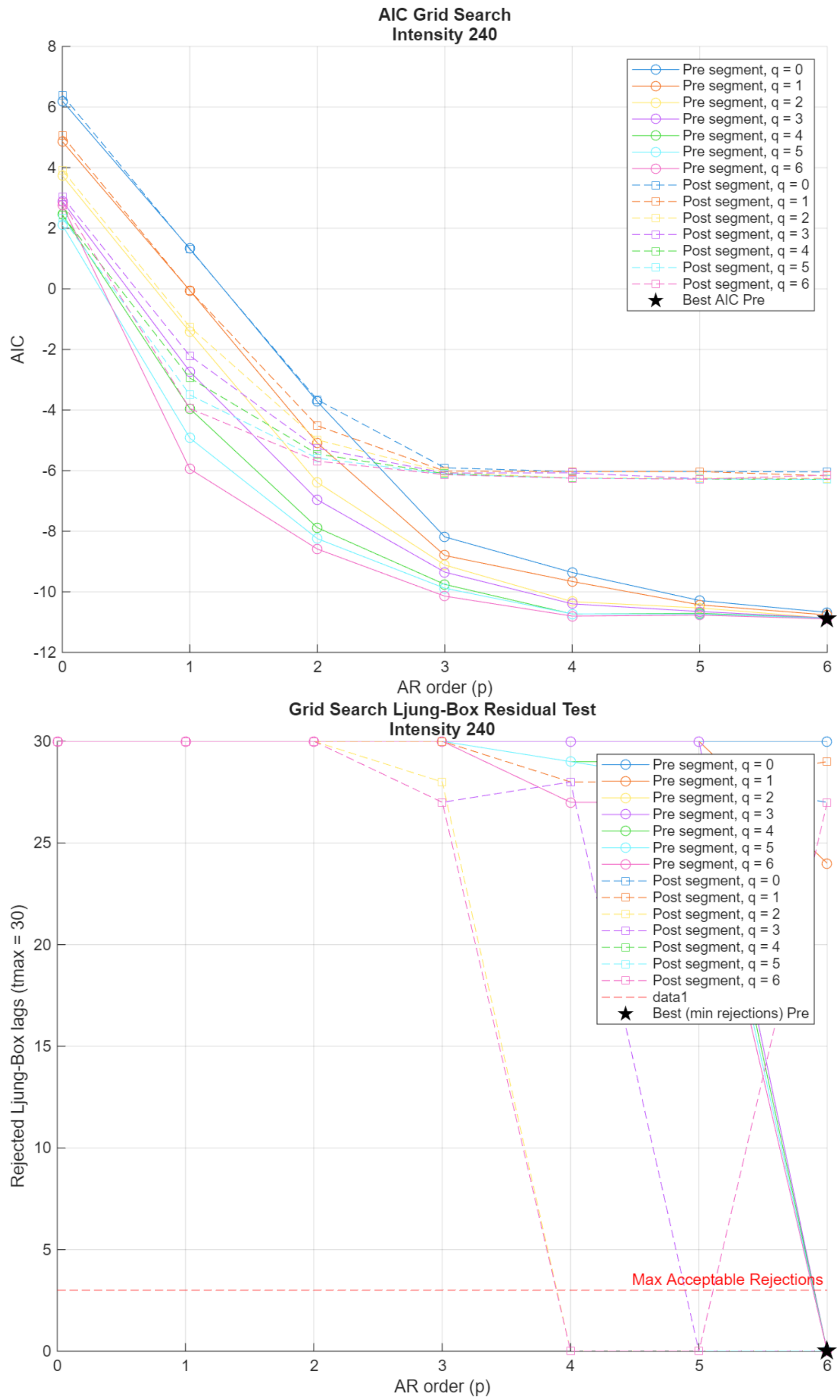

Fig. 5. Grid Search for ARMA Model Selection for EEG recordings for 240% Intensity: The plot illustrates the AIC values for different combinations of AR and MA parameters across the six EEG recordings. The optimal ARMA model is selected based on the lowest AIC value, ensuring a balance between model complexity and fit to the data. The second figure shows the residuals of the fitted ARMA models for each episode, with the Ljung-Box test results indicating whether the residuals exhibit white noise characteristics.

Table 1. Fitted ARMA models, corresponding AIC values, and Ljung–Box test results for each EEG condition.

Fitted Models	**AIC**	**Ljung–Box Test (a=0.05)**
ARMA(4,5) to P120 Pre TMS	−9.03	0/30 lags rejected
ARMA(4,5) to P120 Post TMS	−7.21	0/30 lags rejected
ARMA(3,5) to P180 Pre TMS	−7.20	0/30 lags rejected
ARMA(6,6) to P180 Post TMS	−10.44	2/30 lags rejected
ARMA(6,4) to P240 Pre TMS	−10.87	0/30 lags rejected
ARMA(4,2) to P240 Post TMS	−6.24	0/30 lags rejected

The selected optimal ARMA models were fitted to the training set of each episode, and the following steps were performed:

- The models were trained on the first 600 miliseconds of the episode.
- The models were validated on the last 400 miliseconds to assess its performance.
- Predictions were made from 1 to 10 time steps ahead in order to calculate a robust linear measure.
- The normalized root mean square error (NRMSE) was calculated to evaluate the accuracy of the predictions and this measure is selected to be computed later on for all 120 epochs.

3.4 Linear Measure - NRMSE (Multi-step Prediction)

The selected linear measure for the pilot analysis was the normalized root mean square error (NRMSE) of the ARMA model predictions over multiple forecasting horizons. Specifically, predictions were evaluated for h-step ahead forecasts, where $h = 1, 2, \ldots, 10$. The multi-step NRMSE is defined as:

$$\mathrm{NRMSE}(h) = \sqrt{\frac{\sum_{i=1}^{N-h} (\hat{y}_{i+h} - y_{i+h})^2}{\sum_{i=1}^{N-h} (y_{i+h} - \bar{y})^2}}$$

$$\bar{y} = \frac{1}{N-h} \sum_{i=1}^{N-h} y_{i+h}$$

where:

- y_{i+h} is the true value at time $i + h$,
- $\hat{y}_{i+h}$ is the model's predicted value at time $i + h$, given data up to time i,
- $\bar{y}$ is the mean of the true values over the evaluation window,
- N is the total length of the timeseries.

This formulation allows the assessment of prediction accuracy over different horizons, reflecting the model's ability to generalize beyond immediate next-step forecasting. The pilot analysis considered NRMSE values for each intensity level and pre/post segments, aggregated across $h = 1$ to 10. The resulting NRMSE values across horizons were plotted to observe the degradation of forecast accuracy with increasing h, which is typical in autoregressive models.

The NRMSE provides a normalized and scale-independent measure of the model's predictive accuracy, enabling fair comparison across different time segments and recording conditions.

Importantly, this measure serves as the basis for quantitative statistical comparisons between pre- and post-TMS EEG recordings. By computing NRMSE values for each condition and across forecasting horizons, we can systematically assess how TMS affects the short-term predictability and structure of the EEG signal. Differences in NRMSE values between conditions can then be subjected to paired statistical tests (ANOVA and t-tests) with all the EEG recordings, facilitating inference about changes in linear temporal dynamics induced by stimulation.

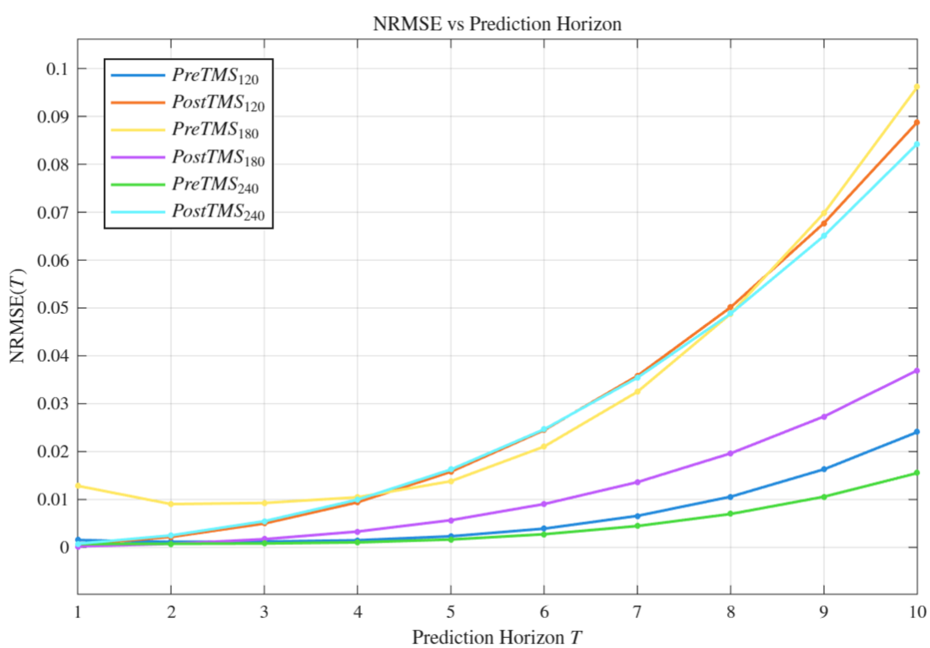

Fig. 6. Normalized Root Mean Square Error (NRMSE) for multi-step predictions across different TMS intensities (120%, 180%, and 240% of LT) and preTMS/postTMS segments. The plot illustrates the prediction accuracy of the ARMA models over multiple forecasting horizons, highlighting the differences in predictability before and after TMS induction.

4 Nonlinear Analysis

For non linear analysis, we focus on the estimation of an invariant nonlinear measure derived from local model prediction accuracy. So, we assume that the underlying system is governed by a deterministic dynamical system, and we aim to reconstruct the state space from scalar time series observations. The pilot analysis is performed on the same six EEG segments that were used for the linear analysis, which were recorded pre- and post-TMS at three different intensities (120%, 180%, and 240% of the motor threshold). The goal is to quantify the predictability of the EEG dynamics by applying local modeling techniques, specifically Local Average Model (LAM) and Local Linear Model (LLM), across all six EEG recordings. The modeling framework involves a systematic exploration of different combinations of reconstruction and forecasting parameters to identify configurations that yield robust and accurate predictions. The model predicts h-step ahead values for each embedded state vector using its k nearest neighbors. These predictions are performed iteratively over multiple lead times to capture both short- and medium-term dynamics. The analysis is divided into four main phases: estimation of embedding parameters, state space reconstruction, local model forecasting, and residual diagnostics.

5 Pilot Architecture for Estimating an Invariant Nonlinear Measure

To quantify nonlinear dynamics in time series, we implement a multi-stage architecture aimed at constructing an invariant nonlinear measure through state-space reconstruction and local model prediction accuracy. This section outlines the methodological pipeline, subdivided into distinct computational and diagnostic phases. To reconstruct the system's dynamics from scalar time series observations, we apply Takens' embedding theorem using the method of delays. This involves determining suitable parameters for the embedding delay (τ) and embedding dimension (m).

5.1 Parameter Search Across Embedding Dimensions, Neighborhood Sizes, and Prediction Horizons

To evaluate the predictability of EEG dynamics, we applied local modeling techniques (LAM/LLM) across all six EEG recordings (pre- and post-TMS at 120%, 180%, and 240% intensities). The modeling framework involved a systematic exploration of different combinations of reconstruction and forecasting parameters to identify configurations that yield robust and accurate predictions. The model predicts h-step ahead values for each embedded state vector using its k nearest neighbors. These predictions are performed iteratively over multiple lead times to capture both short- and medium-term dynamics. For each EEG recording, we varied the embedding dimension m to reconstruct the underlying state space using the method of delays. Candidate values of m (e.g., $2 \leq m \leq 10$) were selected based on preliminary estimates using the False Nearest Neighbors (FNN) algorithm. Concurrently, we explored a range of neighborhood sizes k, ensuring $k > m$ for stability in local linear models. The values of k influence the locality and generalization ability of the models; smaller k focuses on highly localized dynamics, while larger k introduces smoothing but risks diluting nonlinear structures.

Multiple prediction horizons h (e.g., $1 \leq h \leq 10$) were assessed to evaluate the model's performance in short- and mid-range forecasting scenarios. For each (m, k, h) combination with the appropriate τ, we computed the normalized root mean square error (NRMSE) to quantify predictive accuracy.

This grid search was applied independently to each of the six EEG signals, enabling comparative analysis of optimal configurations before and after TMS.

The variation in optimal parameters across conditions reflects differences in dynamical complexity and the potential modulation of neural activity induced by TMS.

Residuals from the best-fitting models were further analyzed for whiteness and normality to confirm model adequacy.

5.2 Phase 1: State Space Reconstruction

The first step involves reconstructing the phase space from a scalar time series using the method of delays. This transformation enables the implicit unfolding of the underlying attractor from the observed dynamics.

Estimation of Time Delay τ. The time delay τ is estimated using the first minimum of the mutual information function, which indicates that the suitable delay is around 15 to 20 lags. The mutual information quantifies the statistical dependence between the time series and its delayed version, providing a robust measure for selecting τ. This is visualized in Figure 7. Additionally, the autocorrelation function (ACF) is computed to verify the delay estimation. The first zero-crossing of the ACF is used as a secondary criterion for τ, ensuring that the delay captures the system's temporal structure without introducing redundancy. The first non-statistical zero-crossing of the ACF is shown in Figure 8 and shows that the delay is around 17 to 24 lags, which is consistent with the mutual information estimation. The selected delay τ is crucial for reconstructing the state space, as it determines the temporal separation between successive points in the embedded phase space. Taking into account the results from both methods, we choose $\tau = 17$ lags for the subsequent analysis.

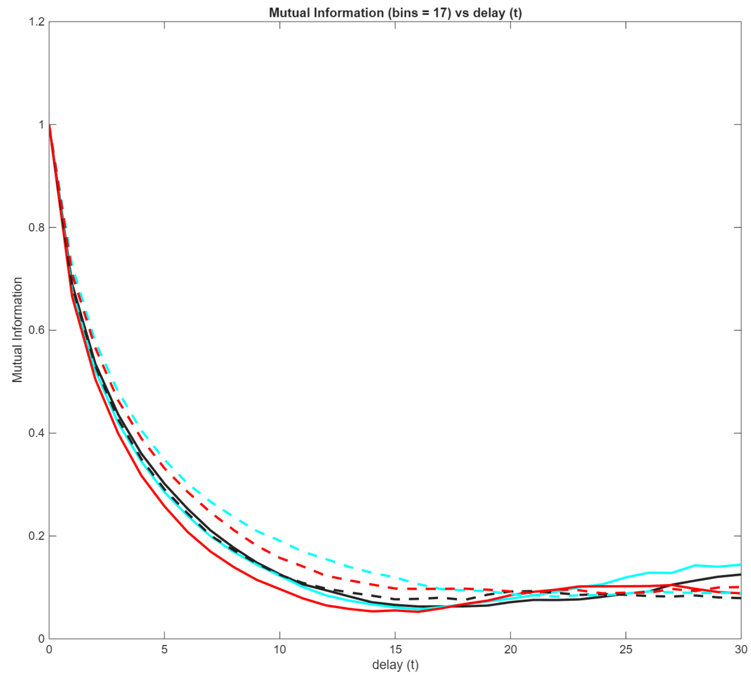

Fig. 7. Estimation of Time Delay τ using Mutual Information. The first minimum of the mutual information function is used to determine the optimal delay for state space reconstruction. The number of bins is set to the rounded integer of $\sqrt{(\frac{N}{5})}$, where N is the length of the time series.

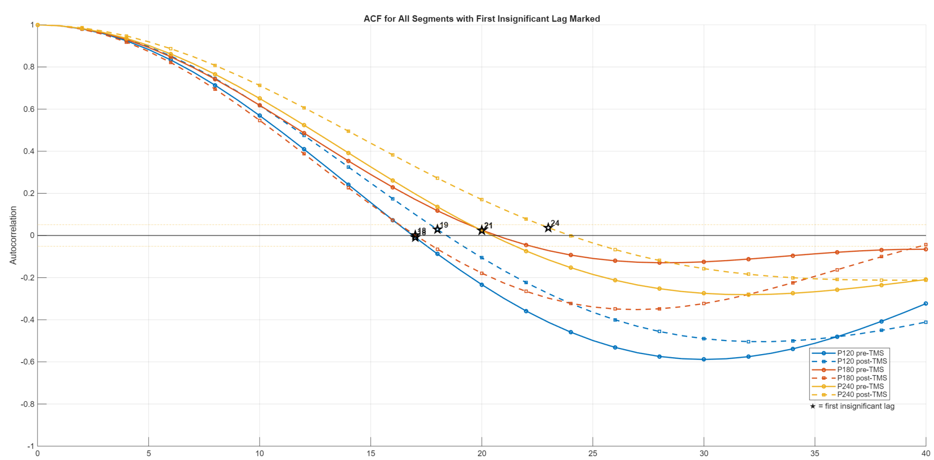

Fig. 8. Estimation of Time Delay τ using ACF. The first zero-crossing of the autocorrelation function is veryfying the above estimation.

Estimation of Embedding Dimension m. The optimal embedding dimension m is determined using the False Nearest Neighbors (FNN) algorithm, which minimizes topological distortion in the reconstructed phase space. It is used as a measure given by the percentage of false nearest neighbors computed for the given range of the delay τ and the embedding dimension m. The figure 9 illustrates the FNN analysis, where the optimal embedding dimension is identified as the point where the percentage of false neighbors drops below 1%. For the six pilot EEG segments, the optimal embedding dimension m is consistently found to be 5, but in the following analysis $m = 4$ was implemented as well to see the difference and maybe achieve sufficient results with smaller m.

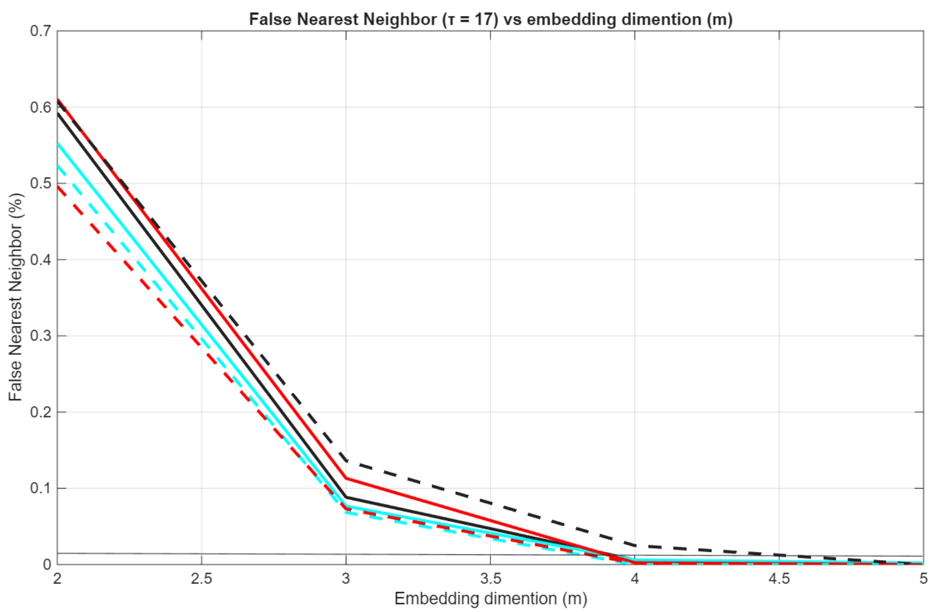

Fig. 9. Estimation of Embedding Dimension m using False Nearest Neighbors (FNN). The optimal embedding dimension is determined by minimizing the percentage of false nearest neighbors, with a threshold of 1% shown by the black line.

The correlation dimension ν is estimated from the scaling of the correlation sum $C(r) \sim r^{\nu}$. To compute this, the time series is first linearly normalized to the interval $[0, 1]$, and $C(r)$ is calculated for radii $r \in (0, 1)$. A log-log plot of $C(r)$ versus r is constructed, and the local slope (i.e., derivative of $\log C(r)$ with respect to $\log r$) is computed by fitting a line to each 5-point window centered on $\log r$. The final estimate of ν is the mean local slope over the interval $[r_1, r_2]$ where the slope variance is minimal, under the constraint that the scale length r_2/r_1 is fixed.[5] The selected Theiler window is set to the default meaning that no constraints are applied to the search of neighbors. Whereas the upper/lower ratio of scaling window is set to 4 meaning that the length of the scaling interval is determined by the factor of 4, $r_2/r_1 = 4$

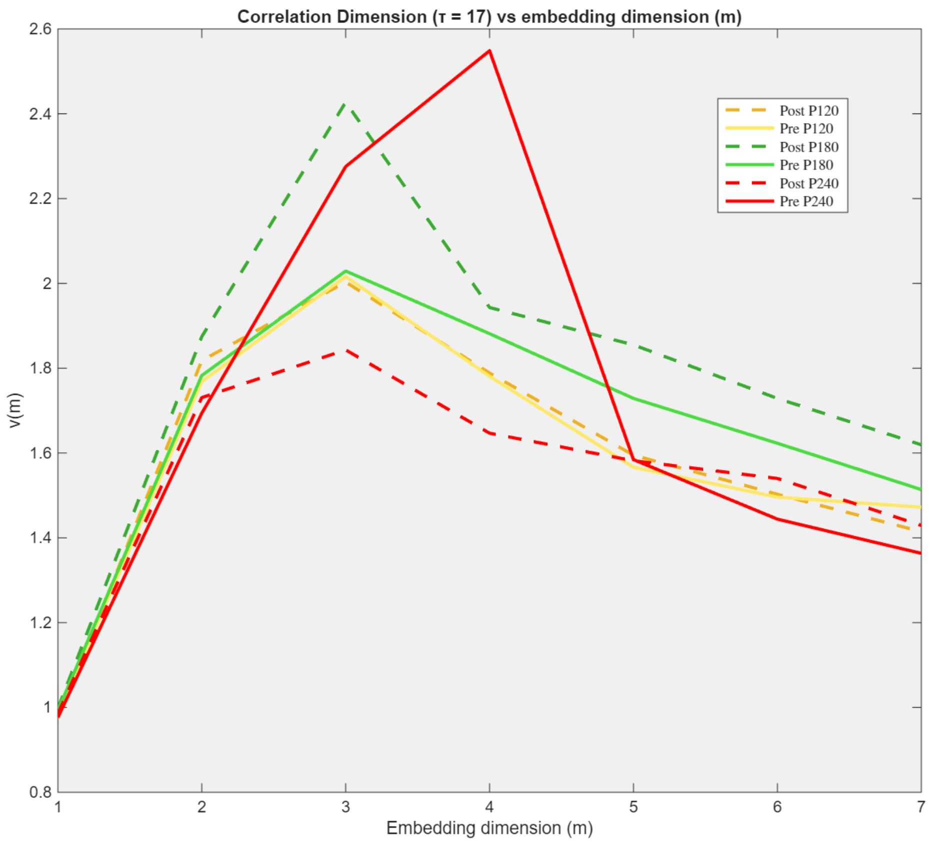

Fig. 10. Correlation dimension $\nu(m)$ as a function of embedding dimension m for pre- and post-stimulation conditions at intesity levels P120, P180, and P240.

The plot of the correlation dimension $\nu(m)$ versus embedding dimension m reveals distinct dynamical behaviors across conditions.10

For all conditions, $\nu(m)$ increases up to $m = 3$ and then saturates or declines, suggesting low-dimensional dynamics. At P120, stimulation induces minimal changes. At P180, $\nu(m)$ increases post-stimulation, indicating higher dynamical complexity. At P240, pre-stimulation values peak the highest but drop post-stimulation, suggesting reduced complexity or system disruption at higher intesity.

An initial increase in $\nu(m)$ from $m = 1$ to $m = 3$ is observed in all curves (6 EEG segments), reflecting the progressive unfolding of the attractor in higher-dimensional spaces.

Beyond $m = 3$, $\nu(m)$ tends to plateau or decrease, particularly in the black and cyan curves, suggesting either the stabilization of the attractor's dimensionality or potential influence of noise and finite data effects at higher embeddings.

The pre-TMS P240% (dotted red) exhibit higher peak values of $\nu(m)$ (up to 2.5), indicating greater complexity or higher-dimensional dynamics.

In contrast, the post-TMS P240 (lower cyan)curves imply simpler, possibly more regular or less chaotic behavior.

Overall, the correlation dimension appears to saturate around $m = 3$–4, implying that the effective attractor dimension resides in this range. The $v(m)$ might be an appropriate invariant measure for the dynamical complexity of the system and might determine the inference of inducing TMS on different intensity levels and what brain functional changes it induces.

5.3 Phase 2: Local Model Forecasting

The above interpretation suggests that the invariant measure derived from local model prediction accuracy will be most informative when using embedding dimensions in this range ($m = 3$–4), as higher dimensions may not yield additional predictive power and could introduce noise. After reconstruction, a local model is fitted to assess the predictability using NRMSE.

Model Type Local Average Model (LAM) and Local Linear Model (LLM) are nonlinear forecasting methods using nearest neighbors in reconstructed state space. LAM averages neighbors (order 0), while LLM fits local linear maps (order 1). We compare local direct and iterative prediction schemes. For $h = 4$, the direct method shows lower NRMSE, suggesting better accuracy in short-term dynamics than iterative prediction.

Two forecasting models were implemented and searched the optimal across various parameters:

- Local Average Model (LAM, model order $q = 0$),
- Local Linear Model (LLM, model order $q \geq m$ or $0 < q < m$ with regularization [6]).

It is used as the optimal one for this occasion to be the local linear direct model with the these parameters:

- Delay: $\tau = 17$
- Embedding Dimension: $m = 3$
- Nearest Neighbors $k = 9$

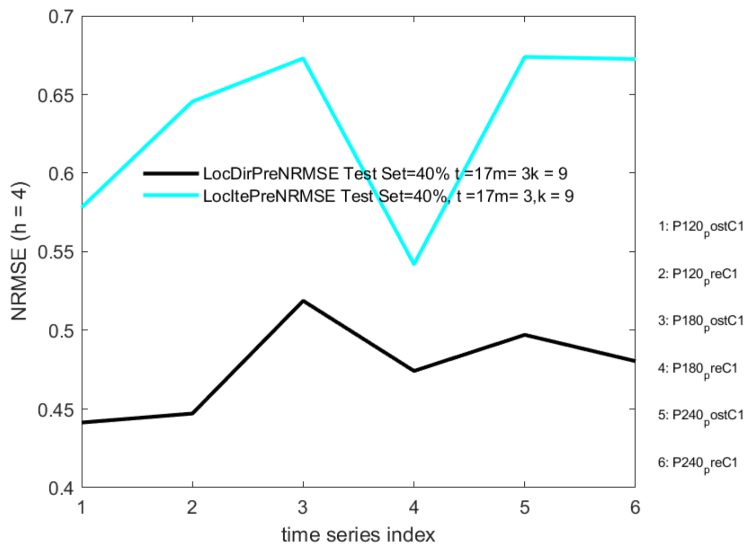

Fig. 11. Comparing NRMSE of Direct and Iterative model for the six pilot EEG recordings for $h = 5$.

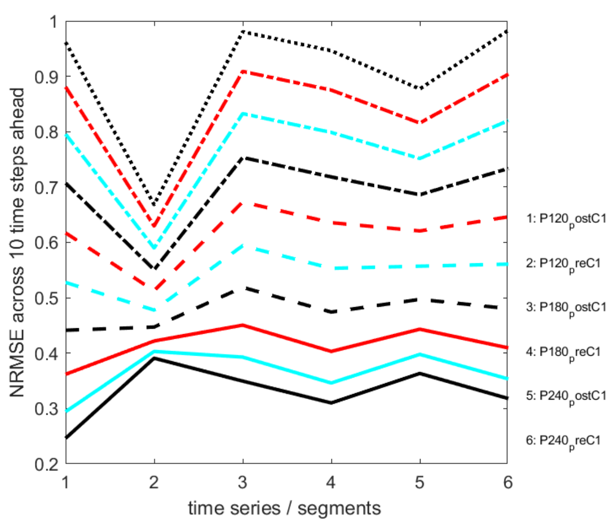

Fig. 12. Checking the prediction time horizon up to 10 steps for the Local Direct Model for the six time series (h varies from 1 the lowest to 10 step the line closer to one.)

Non Linear Measure - NRMSE We selected the Normalized Root Mean Square Error (NRMSE) as the primary metric for evaluating nonlinear model performance due to its robustness in quantifying prediction error relative to the variability of the observed data. In the context of modeling neural or behavioral responses, particularly with local linear models approximating nonlinear dynamics, NRMSE provides a scale-independent measure that facilitates comparison across conditions. This makes it particularly suitable for assessing changes in model accuracy before and after transcranial magnetic stimulation (TMS) across different intensity levels. By normalizing the RMSE to the range or standard deviation of the data, NRMSE ensures comparability across subjects and intensities, allowing us to attribute observed differences to physiological changes rather than measurement scale.

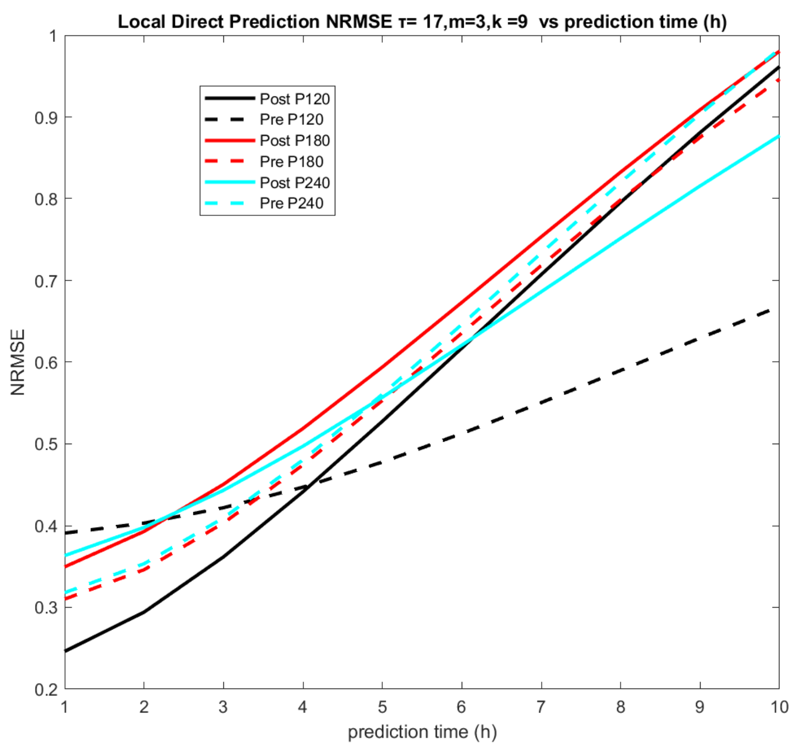

Fig. 13. The performance of Local Linear Direct Prediction Model on Time Horizon h = 10.

To evaluate nonlinear predictability, we computed NRMSE across multiple forecast horizons h using a local direct model. We observed that NRMSE increases with h, reflecting decreasing forecast accuracy. The minimum NRMSE was found at $h = 1$, indicating optimal prediction performance at this timescale. This horizon balances model sensitivity to underlying dynamics with stability in error estimates, making it suitable for statistical comparison across six conditions (pre/post $\times$ 3 intensities).

5.4 Phase 3: Residual Diagnostics

To validate the adequacy of the local prediction models across all categories of epochs, we performed a thorough residual analysis. Specifically, we applied the Portmanteau test (Ljung–Box variant) to assess whether the residuals exhibited white noise characteristics, using a sufficient number of lags to capture potential autocorrelations. In addition to the statistical testing, diagnostic plots of the residuals—including autocorrelation functions and Q-Q plots—were examined to visually inspect the model fit. Furthermore, tests for normality were conducted to evaluate whether the residuals conformed to Gaussian assumptions, thereby supporting the appropriateness and robustness of the modeling framework. The residuals from prediction are analyzed to ensure model adequacy. For the invariant measure to be valid:

- Residuals should exhibit no autocorrelation (white noise),
- Distribution of residuals should be approximately normal.

5.5 Confounders and Sensitivity

Several factors influence the robustness of the invariant measure:

- **Noise:** Additive or dynamical noise can distort attractor geometry.
- **Time Series Length:** Short segments may prevent stable estimation of m, τ, or residuals.
- **Parameter Sensitivity:** Results may vary significantly with changes in k, τ, m, or truncation parameter q.

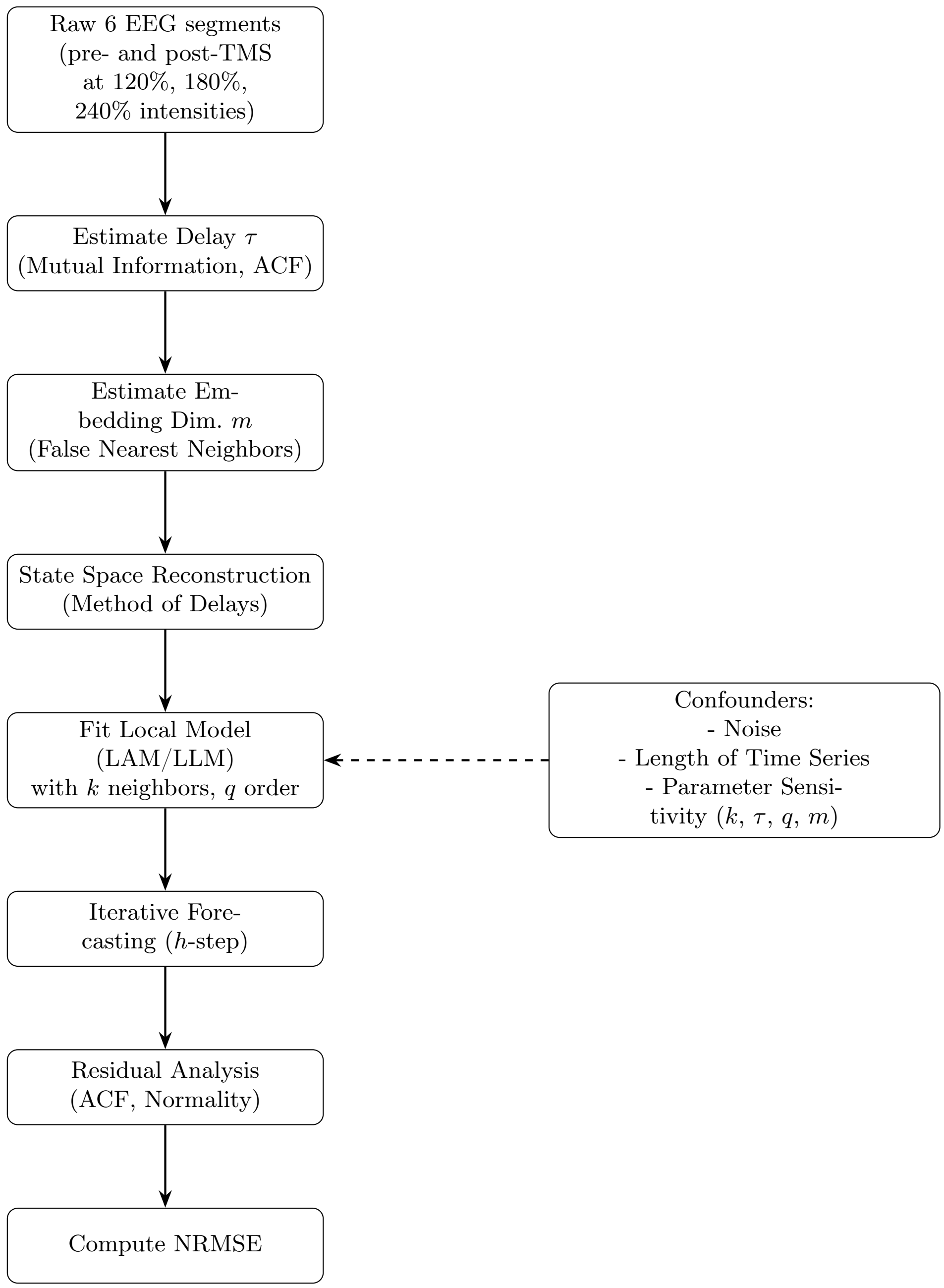

Fig. 14. Methodological Flowchart for estimating an invariant nonlinear measure via reconstruction and local modeling.

6 Statistical Comparison with the selected linear and non-linear measures among all EEG recordings

To assess whether transcranial magnetic stimulation (TMS) induces changes in the underlying brain dynamics, we performed statistical comparisons of the NRMSE distributions across the six experimental conditions. Specifically, pairwise and group-level hypothesis tests (e.g., Wilcoxon signed-rank tests, Mann–Whitney U tests, or ANOVAs, depending on normality and homogeneity assumptions) were conducted on the NRMSE values aggregated over all parameter combinations.

A total of 120 episodes were used for the estimation of the NRMSE linear measure for a prediction horizon of 10 steps ahead, with 20 episodes for each intensity level and then segmented to pre- and post-TMS. This approach allowed for a focused analysis of the effects of TMS on brain activity.

The results were visualized using normality plots to illustrate the distribution of NRMSE values across the different intensity levels and for better understanding of the required assumption for the statistical test.

6.1 Linear Measure - NRMSE

The six optimal ARMA models were trained on the first 600 milliseconds of each episode and validated on the last 400 milliseconds for the pilot study. The normalized root mean square error (NRMSE) was evaluated among all EEG recording depending on the groups. The decision to use NRMSE as a measure of prediction accuracy was based on its ability to provide a normalized error metric, which is particularly useful for comparing model performance across different segments and intensity levels (independent of the scale of the data). The NRMSE values were then summarized for each segment (preTMS and postTMS at each intensity level) using the 10th time step aheadAn argument was raised that the mean or weighted mean or selection on a particular time step NRMSE is a suitable measure for evaluating the prediction accuracy of the ARMA models, as which one will better provide a normalized error metric.

So, we end up with 120 NRMSE values, which were then used to compare the preTMS and postTMS segments across the three intensity levels regarding the four research questions. The table 2 summarizes the NRMSE values for each group, including the number of episodes (n), mean, standard deviation (SD), 25th percentile (P25), median (P50), 75th percentile (P75), skewness, and kurtosis.

6.2 Assumptions for Statistical Tests

To address the research questions, statistical tests were performed on the NRMSE values across the groups. In order to evaluate the ANOVA and t-tests, we need to check of the assumptions of normality, homogeneity of variance and independence for the NRMSE values across the groups.

Table 2. Summary statistics of NRMSE measure computed from linear ARMA models by group

Group	n	Mean	SD	P25	P50	P75	Skewness	Kurtosis
P120 Pre	20	0.0089	0.0031	0.0068	0.0078	0.0112	0.436	-0.924
P120 Post	20	0.0238	0.0129	0.0122	0.0215	0.0317	0.720	0.393
P180 Pre	20	0.0273	0.0064	0.0213	0.0271	0.0315	0.696	-0.269
P180 Post	20	0.0261	0.0143	0.0131	0.0261	0.0350	0.132	-1.230
P240 Pre	20	0.0052	0.0027	0.0036	0.0045	0.0066	1.310	2.370
P240 Post	20	0.0246	0.0100	0.0187	0.0229	0.0282	1.920	6.220

Normality Check via Violin Normality Plots & Shapiro-Wilk Test. After visually inspecting the distributions of NRMSE across the six groups (Figure 15) and performing Shapiro-Wilk tests for normality, we found that only four out of six groups exhibited normally distributed prediction errors. Specifically, the Shapiro test confirmed normality for the P120 Pre, P120 Post, P180 Pre, and P180 Post groups, while the P240 Pre and P240 Post groups deviated from normality. Given this partial violation of the normality assumption, and to ensure a robust group comparison, we proceeded with the non-parametric Kruskal–Wallis test instead of parametric ANOVA.

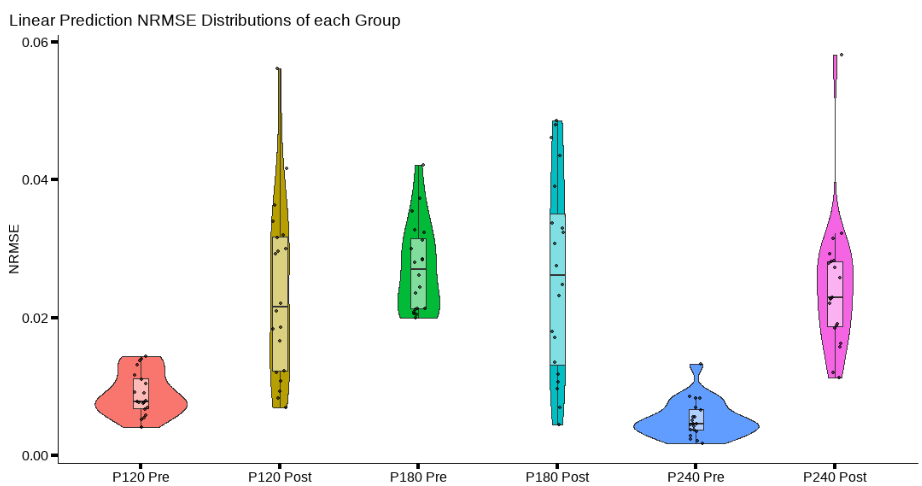

Fig. 15. Violin normality plots of NRMSE values for preTMS and postTMS segments across three TMS intensity levels (120%, 180%, and 240% of LT). The plots illustrate the distribution of NRMSE values, highlighting differences in prediction accuracy before and after TMS administration. The width of each violin indicates the density of NRMSE values at different levels, providing insights into the variability and central tendency of prediction errors across conditions.

Homogeneity of Variance via Levene's Test. Levene's test was conducted to assess the assumption of homogeneity of variances across the six groups. The test returned a highly significant result ($p<0.001$), indicating a clear violation of homoscedasticity. Taken together with the earlier finding that not all groups follow a normal distribution, both key assumptions for parametric ANOVA—normality and equal variances—are not satisfied. Therefore, a non-parametric approach, specifically the Kruskal–Wallis test, is deemed more appropriate for comparing group-level differences in NRMSE.

6.3 Statistical Comparison of Prediction Error

To assess the statistical significance of the differences in ARMA model performance between pre- and post-TMS conditions, we employed both paired t-tests and repeated measures ANOVA on the normalized root mean square error (NRMSE) values computed across 120 EEG time series.

Paired t-Test. For each intensity level, we conducted a two-tailed paired t-test comparing the mean NRMSE values between pre- and post-TMS segments. This test evaluates whether TMS induces a significant change in linear prediction accuracy:

$$H_0 : \mu_{\text{pre}} = \mu_{\text{post}} \quad \text{vs.} \quad H_1 : \mu_{\text{pre}} \neq \mu_{\text{post}}.$$

If the normality assumption was violated (assessed via the Shapiro-Wilk test), the non-parametric Wilcoxon signed-rank test was used as an alternative.

Repeated Measures ANOVA. To evaluate the joint effect of stimulation intensity and TMS condition (pre/post) on prediction accuracy, a two-way repeated measures ANOVA was performed. The within-subject factors were *Segment* (Pre, Post) and *Intensity* (120, 180, 240). This model tested for main effects and interaction:

– Main effect of Segment: difference between pre and post conditions,
– Main effect of Intensity: influence of stimulation level on NRMSE,
– Interaction: whether the effect of TMS differs across intensities.

Post hoc pairwise comparisons were corrected using Bonferroni adjustment.

Interpretation. NRMSE serves as a linear measure of model performance, and its statistical comparison across conditions allows for quantifying the influence of TMS on EEG dynamics. Significant reductions in post-TMS NRMSE suggest improved short-term predictability and potential modulation of linear structure by stimulation.

Statistical Results – Linear NRMSE

- Statistical Tests on NRMSE across the six categories were used to evaluate these research questions:
  1. Differences between preTMS and postTMS for each intensity level.
  2. Differences across intensity levels postTMS.
  3. Differences across intensity levels preTMS.
  4. Differences across intensity levels postTMS, adjusted for preTMS (using the difference between postTMS and preTMS measures).

1. Is there a difference before and after TMS (preTMS vs postTMS) for each stimulation intensity?

- **P120**: Significant difference detected.
  Paired t-test $p = 0.0001$ (Normality $p = 0.5000$)
- **P180**: No significant difference.
  Paired t-test $p = 0.6926$ (Normality $p = 0.2658$)
- **P240**: Significant difference detected.
  Paired t-test $p < 0.0001$ (Normality $p = 0.2426$)

2. Is there a difference between the three stimulation intensities after TMS (postTMS)?

- One-way ANOVA on post-TMS NRMSE yielded:
  $p = 0.8406 \Rightarrow$ **No significant difference** between intensities after TMS.

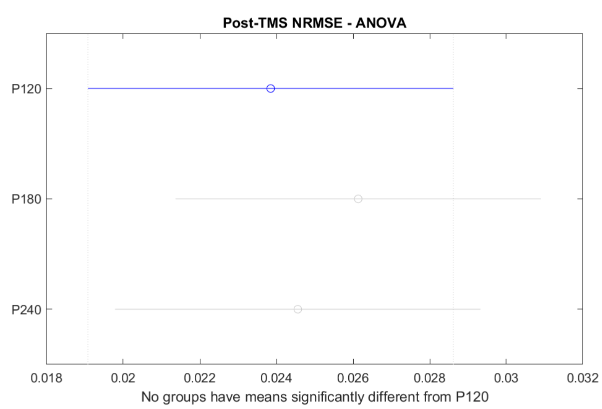

Fig. 16. No significant difference between intensities using linear models.

3. Is there a difference between the three stimulation intensities before TMS (preTMS)?

– One-way ANOVA on pre-TMS NRMSE yielded:
$p < 0.0001 \Rightarrow$ **Significant difference** between intensities before TMS.

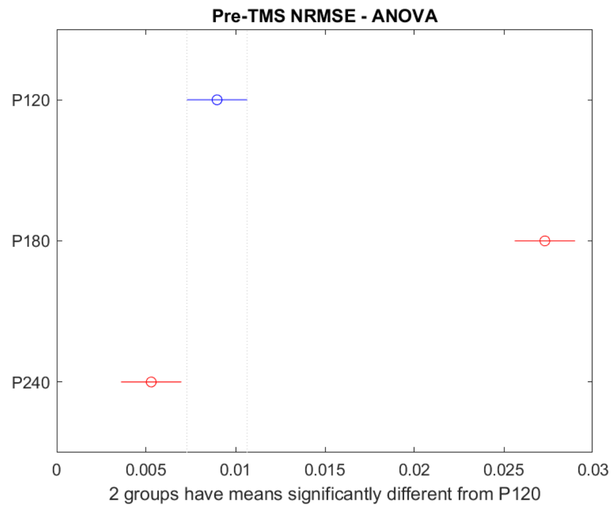

Fig. 17. Pre-TMS between intensities on linear computed NRMSE

4. Is there a difference between the three stimulation intensities in ΔNRMSE (postTMS minus preTMS)?

– One-way ANOVA on ΔNRMSE yielded:
$p < 0.0001 \Rightarrow$ **Significant difference** in the TMS effect across intensities.

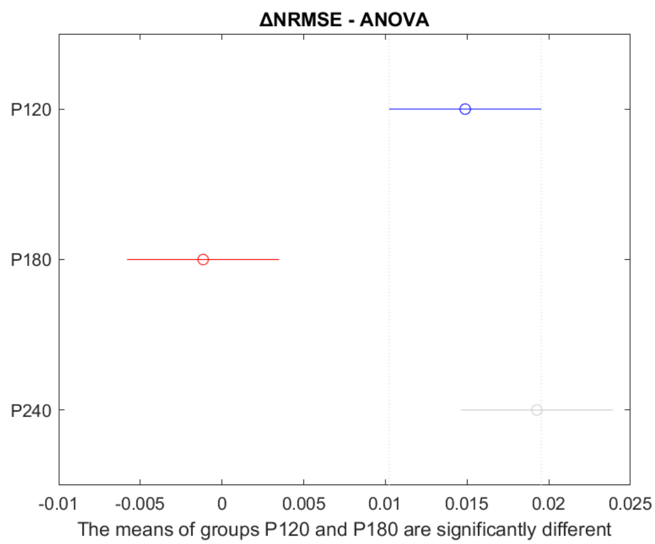

Fig. 18. $\Delta NRMSE$ across stimulation intensities using linear models.

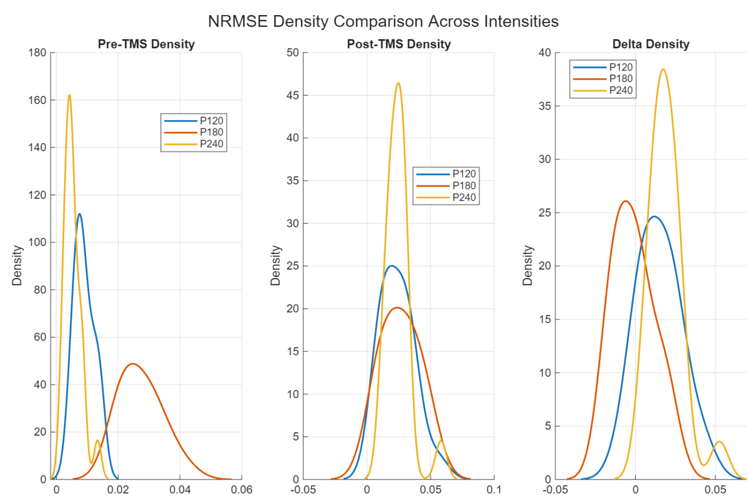

Fig. 19. Boxplot of ΔNRMSE values across stimulation intensities.

6.4 Nonlinear Measure - NRMSE

6.5 Statistical Analysis of Predictive Accuracy Across Conditions

The normalized root mean square error (NRMSE) values computed from the local direct prediction models serve as indicators of predictability within the EEG time series. These values were obtained independently for each recording condition (pre- and post-TMS at 120%, 180%, and 240% intensities), using the optimal configurations for the space reconstruction of embedding dimension $m = 3$, delay $\tau = 17$, neighborhood size $k = 9$, and prediction horizon $h = 1$ to compute the NRMSE. The model search and measure computation were made using the MATS toolbox[3].

A significant difference in predictive accuracy between pre- and post-TMS conditions would suggest that TMS alters the local structure or complexity of the underlying dynamical system, potentially reflecting changes in neural excitability or functional connectivity.

Summary of Assumption Checks and Hypothesis Testing for NRMSE

Data Overview:
NRMSE values were analyzed across six groups (Pre and Post conditions at P120, P180, and P240). Each group included 20 samples. Skewness ranged from 0.199 to 1.66, indicating moderate to high asymmetry in some groups.

Assumption Checks:

- **Independence:** Not violated.
- **Normality:** Evaluated via Shapiro-Wilk test.
  - *Normally distributed:* P120 Pre, P180 Pre, P240 Post.
  - *Not normally distributed:* P120 Post, P180 Post, P240 Pre.
  - Conclusion: Only 3 out of 6 groups passed the normality test.
- **Homoscedasticity:** Assessed using Levene's Test. The assumption was not violated ($p = 0.996 > 0.05$).

Hypothesis Testing:

- Due to non-normality in half of the groups, the non-parametric **Kruskal-Wallis Rank Sum Test** was employed.
- **Null Hypothesis:** All six populations have the same median.
- **Result:** No significant difference observed ($p = 0.941 > 0.05$).
- Post hoc pairwise comparisons were conducted using the Wilcoxon-Mann-Whitney test with Bonferroni correction is recommended.

Conclusion:
Despite variations in skewness and distribution, no statistically significant differences in median NRMSE values were found across the six conditions.

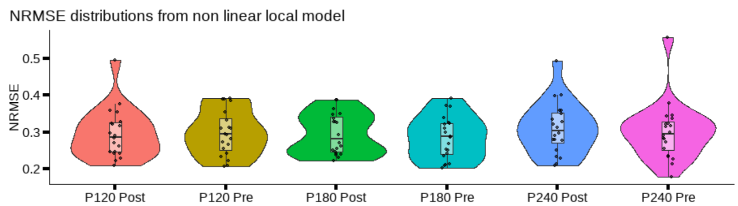

Fig. 20. Shapiro-Wilk normality test results for NRMSE values across six groups. The plot shows the distribution of NRMSE values, indicating which groups are normally distributed and which are not.

6.6 Analysis of NRMSE Results

- Statistical Tests on NRMSE across the six categories were used to evaluate these research questions:
  1. Differences between preTMS and postTMS for each intensity level.
  2. Differences across intensity levels postTMS.
  3. Differences across intensity levels preTMS.
  4. Differences across intensity levels postTMS, adjusted for preTMS (using the difference between postTMS and preTMS measures).

1. **Is there a difference between preTMS and postTMS for each stimulation intensity level?**
   For all intensity levels (**P120**, **P180**, **P240**), normality was assessed using the *Lilliefors test* based on the Kolmogorov–Smirnov test, and normality was not rejected in any case. Therefore, a **paired t-test** was used. The results are:
   - P120: $p = 0.9460$ (Normality $p = 0.5000$)
   - P180: $p = 0.6129$ (Normality $p = 0.2158$)
   - P240: $p = 0.6775$ (Normality $p = 0.3704$)

   Since all p-values are well above 0.05, **there is no statistically significant difference** between preTMS and postTMS for any intensity level.
2. **Is there a difference between the three stimulation intensity levels after TMS (postTMS)?**
   A **one-way ANOVA** was performed on postTMS NRMSE values across the three intensities. The result:
   $$p = 0.6980$$
   Since $p > 0.05$, **there is no statistically significant difference** in NRMSE between the three intensity levels after TMS.
3. **Is there a difference between the three stimulation intensity levels before TMS (preTMS)?**
   Similarly, a **one-way ANOVA** was used for preTMS NRMSE:
   $$p = 0.7558$$

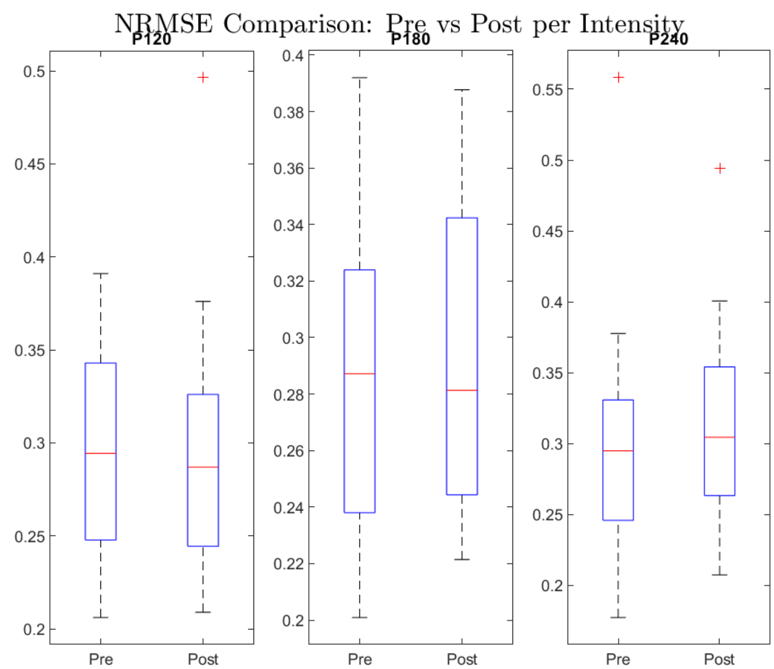

Fig. 21. Boxplots of NRMSE values comparing Pre- and Post-TMS measurements across stimulation intensities (P120, P180, P240). Each pair of boxplots illustrates the distribution of NRMSE values before and after TMS for the corresponding intensity level, showing changes in variability and central tendency.

Again, since $p > 0.05$, **there is no statistically significant difference** between the intensity levels in the preTMS condition. However, it is worth noting that the assumption that preTMS represents a true resting state might be questionable, as the preTMS period starts only 1.5 seconds after the previous TMS, given that TMS pulses are delivered every 2.5 seconds.

4. **Is there a difference between the three intensity levels after TMS, while controlling for preTMS (i.e., using the post-pre difference)?** A **one-way ANOVA** was applied to the difference between postTMS and preTMS NRMSE (ΔNRMSE). The result:

$$p = 0.9114$$

Therefore, **no statistically significant difference** was found in the changes in NRMSE across the three stimulation intensity levels.

The choice of the prediction horizon $h = 1$ plays a crucial role in the calculation of NRMSE, particularly in the context of nonlinear modeling. Using $h = 1$ implies that the model predicts only one step ahead, relying heavily on immediate past values. This short prediction interval may not adequately capture the intrinsic nonlinear dynamics that evolve over longer temporal windows. Consequently, the effects of TMS, which might unfold more gradually or exhibit delayed nonlinear interactions, could remain undetected. This limitation in temporal scope may reduce the sensitivity of the NRMSE metric to changes between pre- and post-TMS conditions, potentially explaining the lack of statistically significant findings in the nonlinear analysis. Employing a longer prediction horizon $h > 1$ could enhance the model's ability to expose these dynamics and lead to more informative and meaningful comparisons or taking into consideration of the mean of multiple prediction steps ahead.

7 Discussion

7.1 Interpretation of Findings

This study investigated how TMS affects EEG dynamics by comparing preTMS and postTMS activity across different stimulation intensities. Both linear (ARMA) and nonlinear (local model) NRMSE measures were analyzed to assess signal predictability and extract a sufficient measure for comparison and intepretation of brain functionality after TMS.

TMS-Induced Effects. In the linear analysis, significant increases in NRMSE were found after TMS at 120% and 240% intensities. This suggests a disruption in short-term predictability, indicating that TMS can affect the linear structure of EEG signals.

In contrast, the nonlinear models showed no significant differences between preTMS and postTMS across all intensities. This result implies that local nonlinear dynamics were not strongly affected—at least with the current setup using a prediction horizon of $h = 1$.

Intensity-Dependent Effects. No consistent intensity-dependent effects were observed postTMS. PostTMS NRMSE values did not significantly differ across the three intensities in either the linear or nonlinear model. However, differences in preTMS NRMSE were observed in the linear case, likely reflecting natural baseline variability rather than systematic intensity-related changes.

Implications for the PreTMS Baseline. The validity of the preTMS segment as a true resting baseline is questionable. TMS pulses were delivered every 2.5 seconds, and the preTMS window began only 1.5 seconds after the previous pulse. It is possible that residual effects from prior stimulation influenced the preTMS segment, especially at higher intensities. This may reduce the contrast between preTMS and postTMS activity and obscure subtle dynamic changes.

On the Sensitivity of Prediction Horizon. The nonlinear NRMSE analysis used a short prediction horizon ($h = 1$), which limits sensitivity to longer-range or delayed dynamic effects. TMS-induced changes may unfold over multiple steps, and one-step predictions might not capture these temporal structures. Using longer prediction horizons or averaging over multiple steps could provide greater sensitivity in detecting nonlinear effects.

Overall Implications. These findings suggest that linear models are more sensitive to short-term changes in EEG predictability induced by TMS, particularly at moderate to high intensities. While nonlinear models are theoretically more expressive, they may require more tailored parameters to detect subtle changes.

Future studies should explore varying prediction horizons and ensure that preTMS segments are temporally distant from preceding TMS pulses to improve baseline accuracy. These refinements could enhance the sensitivity of both linear and nonlinear methods in capturing TMS-induced changes in brain dynamics.

Comment on the Pre-TMS Resting State Assumption

The assumption that the pre-TMS condition represents a resting or baseline state may not be valid in the current experimental setup. In this protocol, pre-TMS measurements are obtained only 1.5 seconds after the previous TMS pulse, while the pulses are spaced at 2.5-second intervals.

However, TMS is known to induce **neural aftereffects** that can persist for several seconds, depending on the stimulation parameters such as intensity, frequency, and cortical region. Consequently, the brain may not have returned to a true baseline state within 1.5 seconds. Thus, what is referred to as "pre-TMS" may still carry residual effects from the previous TMS pulse, and not represent a genuine resting condition.[7]

This observation has important implications:

- The computed ΔNRMSE (post minus pre) values may **underestimate the actual TMS-induced effects**, since the "pre" measurement may already be altered.

– This could also explain the **lack of statistically significant differences** between pre- and post-TMS measures, as both might reflect stimulated brain states to some extent.

Recommendation: Future studies should consider incorporating longer intervals before TMS pulses or introducing dedicated resting-state measurements well-separated from any preceding TMS activity. This would provide a more reliable baseline for comparison and strengthen the validity of pre-post statistical analyses.

7.2 Limitations

– Focus on a single EEG channel may limit generalizability.
– Assumption of stationarity in EEG segments.

8 Conclusion

The study provides insights into the effects of TMS on brain activity, highlighting potential intensity-dependent disruptions. Future work could expand the analysis to multiple channels and incorporate additional nonlinear measures.[8]

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A Appendix

A.1 Code Availability

- Description of MATLAB scripts used for analysis to be submitted separately (zip file) and the R script for the statistical assumptions[4].
- The MATS toolbox was used for nonlinear analysis, available at `https://eeganalysis.web.auth.gr/`.